

Course Syllabus

General Information

Faculty	Information Technology		
Department	Software Engineering		
Semester	2nd	Academic Year	2025/2026

Course Details

Course Title	Software Development and Documentation			Course No.		0423502	
Type of Learning	☐ On- campus ☒ Blended (..... Platform) ☐ Online (..... Platform)						
Credit Hours	3	Theoretical	3	Practical	0		
Course Level (according to JNQF)			7	JNQF Hours*		12.1	
Pre-requisite	0413402			Co-requisite			

Course Instructor Information

Instructor Name	Dr. Ahmed Hassan		
Instructor E-mail	a.hassan@meu.edu.jo		
Course Time	12- 1 (Sun, Tues, Thurs)		
Office Hours	10-11 Sun, Tues, Thurs and 12:30-2 Mon, Wed		
Office Number	N132	Office Phone	
Lab Instructor Name –If assigned-			

Sources and References

1. Textbook
<i>Docs for Developers: An Engineer's Field Guide to Technical Writing</i> , Jared, Apress, 10 th edition, 2021
2. References
1. Clements et al. <i>Documenting Software Architecture, Views and Beyond</i> , 2nd Edition (Addison-Wesley Professional: 2010).
2. Thomas T. Barker <i>Writing a Software Documentation: a Task-oriented approach</i> (Pearson Education: 2003).
3. Sommerville <i>Software Documentation</i> , (Lancaster University: 2003)

Teaching Methods

- | | | |
|-----------------------------------|----------------------------|---------------|
| 1. Lectures | 2. Visualization Tools | 3. Discussion |
| 4. Projects and Students Analysis | 5. Quizzes and Assessments | |

Course Description

This course focuses on the principles and practices of creating effective technical documentation for software and related technologies. Students will learn how to identify audience needs, plan and structure documentation, draft and edit content for clarity and usability, integrate code samples, gather and apply user feedback, and evaluate documentation quality using appropriate metrics. Emphasis is placed on developing practical skills through hands-on projects that result in clear, accurate, and user-centered documentation.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	Identify user needs and apply documentation techniques to create clear, accurate, and effective technical content.	PLO1	K1-I		
2	Explain the processes of audience analysis, documentation planning, and information design, highlighting how these elements contribute to effective and user-centered documentation.	PLO1	K1-P		
3	Follow best practices and use appropriate tools to create, improve, and evaluate technical documentation.	PLO6		S2-P	
4	Apply knowledge and skills to plan, develop, and present a comprehensive documentation project that integrates audience analysis, drafting, editing, tool usage, and publishing workflows.	PLO6		S2-D	

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture Type	Lecture No.		Assessment Tool	Summative/ Formative
1	Course overview; role of documentation	Lecture (1)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	1	Discussion & Questions	Formative
2	Ch.1 Understanding your audience (users, goals, needs) -Audience research methods (existing/new data)	Lecture (1)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	1		Summative
3	Ch.1 Understanding your audience -User journey maps & friction logs	Lecture (1)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	1	Discussion & Questions	Formative
4	Ch.2 – Planning Your Documentation Content types: code comments, READMEs, conceptual, procedural, reference	Lecture (1)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	1	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	1	Discussion & Questions	Formative
5	Ch.3: Drafting Documentation: titles, outlines, headers, paragraphs -Writing setup: tools, writing goals, workflows	Lecture (1)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	2	Discussion & Questions	Formative
6	Ch.3: Drafting Documentation: skimmable writing (clarity, formatting) -Overcoming writer's block, using templates	Lecture (1)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	2		Summative
7	Ch. 4: Editing Documentation: accuracy, completeness, structure -Reviews: peer review, technical review, integrating feedback	Lecture (1)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	2	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	2	Discussion & Questions	Formative
8	Mid Exam					Summative
9	-Ch.5 – Code samples: types & principles (clear, concise, usable) -Designing & testing code samples	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3	Discussion & Questions	Formative
10	-Ch.5 – Code samples: types & principles (clear, concise, usable)	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative

	-Designing & testing code samples	Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3		Summative
11	Ch.8 – Gathering and integrating feedback (channels, surveys, councils)	Lecture (1)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	3	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	3		Summative
12	Ch.8 – Gathering and integrating feedback -Triaging feedback & integrating improvements	Lecture (1)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	4	Discussion & Questions	Formative
13	Ch.9-Measuring Documentation Quality	Lecture (1)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	4		Summative
14	Ch.9-Measuring Documentation Quality -Migrating & maintaining documentation structures	Lecture (1)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (2)	Lecture On- campus	4	Discussion & Questions	Formative
		Lecture (3)	Lecture On- campus	4		Summative
15	Project presentation	Lecture (1)	Lecture On- campus	5	Discussion	Summative
		Lecture (2)	Lecture On- campus	5	& Questions	Summative
		Lecture (3)	Lecture On- campus	5	Discussion & Questions	Formative
16	Final Exam	Lecture (1)	Lecture On- campus	1-5		Summative

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Mid Exam	25%	Week 8
Project	35%	Week 15
Final Exam	40%	Week 16

Course Policies

Item	Policy
Quizzes	- No makeup quizzes. - A minimum of 2 quizzes are given. - Each Quiz is out of 10.

	If more quizzes are given, then the lowest quiz's grade is dropped.
Exams	The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	Makeup exam should not be given unless there is a valid excuse.
Drop Date	The last day to drop the course is according to the University regulations.
Academic Honesty	- Cheating or copying on exams or quizzes is an illegal and unethical activity. - Standard MEU policy will be applied. - All graded assignments must be your own work (your own words).
Attendance	- Excellent attendance is expected. - MEU policy requires the faculty member to assign ZERO grade if a student misses 20% of the classes. - Sign-in sheets will be circulated. - If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	The student should expect to spend 6 hours per week as an average workload.
Graded Exams	Instructor should return exam papers graded to students within one week after the exam date.
Participation	- Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any questions. - Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	- Assignments are as mentioned in the previous table. - Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. - More details about phases and topics of the projects will be given later on.
Others	

Rubrics and Feedbacks (If Applicable)

Quiz and Assignment and Exams will be (Key solutions based).

Project will be on research topic related to security chosen by the student from a list of Topics, followed by a 10-minute presentation. This rubric will be considered for the project and presentation.

Poor	Adequate	Very Good	Excellent	Criteria
Student does not understand the project requirements and goals (1 mark)	Student demonstrates a basic understanding of the project requirements and goals. (2 marks)	Student shows a good understanding of the project requirements and goals. (3 marks)	Student demonstrates a deep understanding of the project requirements and goals. (4 marks)	Project Understanding
Student lacks the necessary technical skills for the project. (1 mark)	Student demonstrates basic technical skills but lacks depth or	Student shows strong technical skills and applies them proficiently. (2	Student demonstrates exceptional technical skills and	Technical Skills

	efficiency. (1.5 marks)	marks)	utilizes advanced techniques effectively. (3 marks)	
Student's project is non-functional and riddled with critical issues. (1 mark)	Student's project has basic functionality but is plagued by performance issues or major bugs. (2 marks)	Student builds a project with good functionality and performance, with minor issues. (3 marks)	Student develops a project with excellent functionality and performance, free from any issues or bugs. (4 marks)	Functionality and Performance
Student fails to provide any meaningful documentation. (1 marks)	Student provides basic documentation but lacks detail and organization. (2 marks)	Student offers adequate documentation but could improve organization and clarity. (3 marks)	Student provides comprehensive and well-organized documentation, making it easy for others to understand and contribute to the project. (4 marks)	Documentation

* JNQF Hours Calculation

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	2	14	28
Tutorials			
laboratory	2	14	28
Activities for Assessment and Evaluation			
First Exam/ Midterm	2	1	2
Second Exam (If applicable)			
Final Exam	2	2	4
Quizzes	2	0.5	1
Self-Learning Activities			
Assignments/ Homework	1	10	10
Case Studies			
Presentations	1.5	2	3
E-learning			
Extra Readings	4	10	10
Total Hours	121		
JNQF Hours	Total Hours/10 = 12.1		